

Лабораторная работа №4

Продвинутое использование git

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РУДН

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Цель работы

- Получение навыков правильной работы с репозиториями git
- Освоение рабочего процесса Gitflow Workflow
- Изучение семантического версионирования и Conventional Commits
- Настройка автоматического создания CHANGELOG.md

Установка ПО

- Установка **git-flow**: `sudo apt install git-flow`
- Установка **Node.js** и **npm**: `sudo apt install nodejs npm`
- Установка **pnpm**: `sudo npm install -g pnpm`
- Установка **commitizen** и **standard-changelog**: `pnpm add -g commitizen standard-changelog`

```
(base) hhl@LAPTOP-Q58L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ sudo apt update
[sudo] password for hhl:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Hit:5 https://cli.github.com/packages stable InRelease
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1504 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1806 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [241 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.1 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [975 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.1 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [20.6 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:15 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [332 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.7 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1564 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [318 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [32.9 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:25 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7300 B]
Get:26 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
```

Создание репозитория

- **Создание на GitHub:** репозиторий git-extended
- **Клонирование и первый коммит:**

```
git init
git remote add origin git@github.com:haoczan2/git-ext
echo "# Git Extended Lab" > README.md
git add README.md
git commit -m "first commit"
git push -u origin master
```

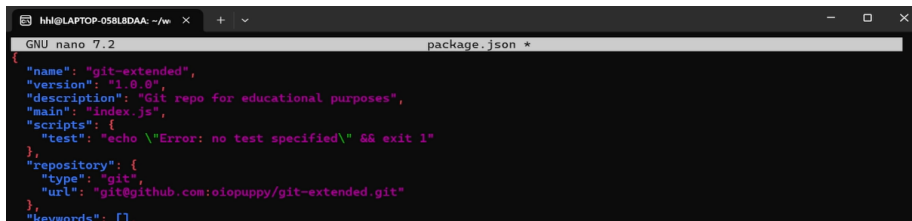
```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git init
Initialized empty Git repository in /home/hhl/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended
/.git/
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git remote add
origin git@github.com:oiopuppy/git-extended.git
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ echo "# Git Ext
ended Lab" > README.md
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git add README.
md
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git commit -m "
first commit"
[master (root-commit) a9cbad7] first commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git branch -M m
aster
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git push -u ori
```

Настройка Conventional Commits

- Инициализация **pnpm**: `pnpm init`
- Настройка **commitizen** в `package.json`:

```
"config": {  
  "commitizen": {  
    "path": "cz-conventional-changelog"  
  }  
}
```

- Первый **conventional commit**: `git cz` (тип: chore)

A screenshot of a terminal window with a dark background. The title bar shows 'hhl@LAPTOP-0581BDAA: ~/w' and window control buttons. The editor is GNU nano 7.2, editing package.json. The JSON content is as follows:

```
{  
  "name": "git-extended",  
  "version": "1.0.0",  
  "description": "Git repo for educational purposes",  
  "main": "index.js",  
  "scripts": {  
    "test": "echo \\\"Error: no test specified\\\" && exit 1"  
  },  
  "repository": {  
    "type": "git",  
    "url": "git@github.com:oiopuppy/git-extended.git"  
  },  
  "keywords": []  
}
```

Инициализация Gitflow

- **Запуск git-flow:** `git flow init`
 - Ветка релизов: `master`
 - Ветка разработки: `develop`
 - Префиксы: `feature/`, `release/`, `hotfix/`
 - Префикс тегов: `v`
- **Отправка веток:** `git push --all`

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git flow init
Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [/home/hhl/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended/.git/hooks]
```

Рисунок 4: Gitflow init

Создание релиза v1.0.0

- **Релизная ветка:** `git flow release start 1.0.0`
- **Генерация CHANGELOG:** `standard-changelog --first-release`
- **Завершение релиза:** `git flow release finish 1.0.0`
- **Отправка тегов:** `git push --tags`
- **Релиз на GitHub:** `gh release create v1.0.0 -F CHANGELOG.md`

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ gh release create v1.0.0 -F CHANGELOG.md
https://github.com/oioopuppy/git-extended/releases/tag/v1.0.0
```

Рисунок 5: Релиз на GitHub

Разработка новой функциональности

- **Feature ветка:** git flow feature start feature_branch
- **Добавление файла:** echo "console.log('new feature');" > new-feature.js
- **Коммит с типом feat:** git cz
- **Завершение:** git flow feature finish feature_branch

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0
? Select the type of change that you're committing: feat:      A new feature
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 94 chars):
  (20) add new feature file
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[feature/feature_branch 5dc2f7a] feat: add new feature file
1 file changed, 1 insertion(+)
create mode 100644 bew-feature.js
```

Рисунок 6: Feature commit

Создание релиза v1.2.3

- **Обновление версии** в package.json: 1.0.0 → 1.2.3
- **Обновление CHANGELOG**: standard-changelog
- **Завершение релиза**: git flow release finish 1.2.3
- **Отправка**: git push --all && git push --tags
- **Релиз на GitHub**: gh release create v1.2.3 -F CHANGELOG.md

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ gh release create v1.2.3 -F CHANGELOG.md
https://github.com/oioopuppy/git-extended/releases/tag/v1.2.3
```

Рисунок 7: Релиз v1.2.3

- **Граф коммитов:** `git log --oneline --graph --all`

```
(base) hhl@LAPTOP-058L8DAA:~/work/study/2026-2027/Операционные системы/os-intro/labs/lab04/git-extended$ git log --oneli
ne --graph --all
*   b585064 (HEAD -> develop, origin/develop) Merge tag 'v1.2.3' into develop
/\
*   49c3cca (tag: v1.2.3, origin/master, master) Merge branch 'release/1.2.3'
/*
*   8a968a5 chore(site): update changelog for 1.2.3
/\
*   425fb2c Merge tag 'v1.0.0' into develop
/\
*   c4907da (tag: v1.0.0) Merge branch 'release/1.0.0'
/*
*   c3c5a99 chore(site): add changelog
/\
*   5dc2f7a feat: add new feature file
/\
*   7749a59 chore: initialize package.json and commitizen config
*   a9cbad7 first commit
```

Рисунок 8: Git log

- Освоен рабочий процесс **Gitflow Workflow**
- Изучено **семантическое версионирование** (SemVer)
- Применены **Conventional Commits** для стандартизации коммитов
- Настроена автоматическая генерация **CHANGELOG.md**
- Освоена работа с **тегами** и **релизами** на GitHub
- Получен практический опыт работы с **git-flow**, **commitizen**, **standard-changelog**

Спасибо за внимание!